

The project is based on an optimization model:

$\max \sum_i w_i r_i$ Here r_i represents the expected daily return of stock i , calculated from the mean of historical log returns. I used log returns because they are additive over time, which makes them more suitable for working with covariances and optimization than simple returns.

$\sum w_i = 1$ (This constraint ensures all (planned) capital is invested)

$w^T \Sigma w \leq Rmax$ (This constraint enforces a maximum acceptable level of variance)

$0 \leq w_i \leq 0.325$ (Portfolio is diversified, no more than 32,5% of the investment goes into one stock)

I chose to maximize expected return for a given risk level rather than minimizing variance, because I want the best possible return the optimizer can find within each risk budget.

Risk model:

For risk, I used the daily variance that we get from volatility:

$$\sigma_{daily} = \frac{\sigma_{annual}}{\sqrt{252}}$$

σ_{daily}^2 is the variance.

I compute the covariance matrix to capture how asset returns move with respect to each other.

The portfolio variance is computed as

$$w^T \Sigma w = \sum_i \sum_j w_i w_j Cov(r_i, r_j)$$

The variance term $w^T \Sigma w$ captures not only individual asset risk, but also how assets interact. Assets that move in opposite directions reduce overall portfolio risk through diversification.

Variance range:

For a given list of stocks, we find the highest and lowest possible risks so that, later on, we can check whether our risk strategies are within the range.

There are 2 functions – `max_variance`, `min_variance` – that find weights which maximize/minimize our variance under the weight constraints that I used for returns (since we need the same weights in both models).

I define target annual volatility levels 12%, 18%, 28% for conservative, balanced, and aggressive, respectively. After converting them to daily variance with the `annual_vol_to_daily_var` function, the values are 0.000057, 0.000129, and 0.000311. They are then checked to see whether they

are in range for the particular portfolio (list of chosen stocks). If not, we use the max/min variance instead.

More conservative risk levels lead to portfolios with lower variance but also lower expected returns, while higher risk levels allow the optimizer to allocate more weight to volatile assets, increasing potential return.

On the next page, you can find some of the results the program gives when running for a few lists of stocks. I chose a high-volatility list to show the lower bound binding, a low-volatility list to show the upper bound binding, and a mixed list to demonstrate the optimizer's behavior when both extremes are available.

Disclaimer:

Returns are computed from 10 years of historical data covering a prolonged bull market. The optimizer selects weights that performed well historically, which structurally biases return estimates upward. These are not forward-looking projections.

For the following list of stocks: AAPL, MSFT, GOOGL, AMZN, KO, TLT, GLD, INTC, NVDA

Feasible variance range: [0.000039, 0.000358]

conservative: 0.000057 (in range)

balanced: 0.000129 (in range)

aggressive: 0.000311 (in range)

CONSERVATIVE — Return: 15.04% | Volatility: 12.00%

GOOGL: 32.50%

TLT: 22.08%

NVDA: 20.19%

INTC: 10.52%

AAPL: 8.53%

AMZN: 4.71%

GLD: 1.48%

Not included: MSFT, KO

BALANCED — Return: 24.00% | Volatility: 18.00%

GOOGL: 32.50%

INTC: 23.50%

TLT: 21.23%

AAPL: 14.65%

AMZN: 6.42%

NVDA: 1.70%

Not included: GLD, MSFT, KO

AGGRESSIVE — Return: 33.41% | Volatility: 28.00%

INTC: 32.50%

AAPL: 32.50%

AMZN: 25.27%

GOOGL: 9.73%

Not included: GLD, MSFT, TLT, KO, NVDA

List of stocks with low volatility (last 10 years): KO, PG, JNJ, TLT, GLD, VZ, CL

Feasible variance range: [0.000035, 0.000059]

conservative: 0.000057 (in range)

balanced: 0.000059 (not in range, used max variance)

aggressive: 0.000059 (not in range, used max variance)

CONSERVATIVE — Return: 10.16% | Volatility: 12.00%

PG: 32.50%

JNJ: 32.49%

TLT: 22.71%

GLD: 11.91%

VZ: 0.38%

Not included: CL, KO

BALANCED — Return: 10.23% | Volatility: 12.18%

PG: 32.50%

JNJ: 32.50%

TLT: 32.50%

GLD: 2.50%

Not included: CL, KO, VZ

AGGRESSIVE — Return: 10.23% | Volatility: 12.18%

PG: 32.50%

JNJ: 32.50%

TLT: 32.50%

GLD: 2.50%

Not included: CL, KO, VZ

We can see that in the list of low volatility stocks, 18% and 28% volatility cannot be achieved, so the program uses the maximum possible variance of 12.18% for both the balanced and aggressive strategies. For these stocks, we get low risk and low returns.

High return stocks (last 10 years): NVDA, AAPL, MSFT, AMZN, AVGO, TSLA

Feasible variance range: [0.000254, 0.000610]

conservative: 0.000254 (not in range, used min variance)

balanced: 0.000254 (not in range, used min variance)

aggressive: 0.000311 (in range)

CONSERVATIVE — Return: 24.75% | Volatility: 25.32%

AMZN: 32.50%

NVDA: 32.50%

AAPL: 24.60%

MSFT: 10.40%

Not included: TSLA, AVGO

BALANCED — Return: 24.75% | Volatility: 25.32%

AMZN: 32.50%

NVDA: 32.50%

AAPL: 24.60%

MSFT: 10.40%

Not included: TSLA, AVGO

AGGRESSIVE — Return: 31.74% | Volatility: 28.00%

NVDA: 32.50%

AMZN: 29.90%

MSFT: 19.91%

AVGO: 16.69%

AAPL: 1.00%

Not included: TSLA

For the high-return stock list, we see the opposite: 12% and 18% cannot be achieved, so the program uses the minimum variance of 25.32%. In this case, we get high risk and high returns.

Mixed from both lists: AAPL, KO, GLD, TLT, TSLA, NVDA, JNJ

Feasible variance range: [0.000036, 0.000518]

conservative: 0.000057 (in range)

balanced: 0.000129 (in range)

aggressive: 0.000311 (in range)

CONSERVATIVE — Return: 15.74% | Volatility: 12.00%

KO: 32.50%

GLD: 19.42%

NVDA: 15.40%

TSLA: 12.53%

TLT: 11.65%

AAPL: 8.31%

JNJ: 0.19%

BALANCED — Return: 24.43% | Volatility: 18.00%

KO: 32.50%

TSLA: 25.90%

GLD: 21.38%

AAPL: 13.89%

TLT: 6.12%

JNJ: 0.21%

Not included: NVDA

AGGRESSIVE — Return: 33.41% | Volatility: 28.00%

TSLA: 32.50%

AAPL: 32.50%

KO: 20.97%

JNJ: 14.03%

Not included: GLD, TLT, NVDA

In the mixed list, the behavior is different. For the conservative strategy, the optimizer mainly chooses stocks from the low volatility group, and for the aggressive strategy it includes more high volatility stocks. We even get a higher return than in the high-return-only list (for aggressive strategy), because of the risk constraint and diversification. This shows that diversification does not only reduce risk, but can also improve returns under a risk constraint, by allowing the optimizer to combine assets more efficiently.